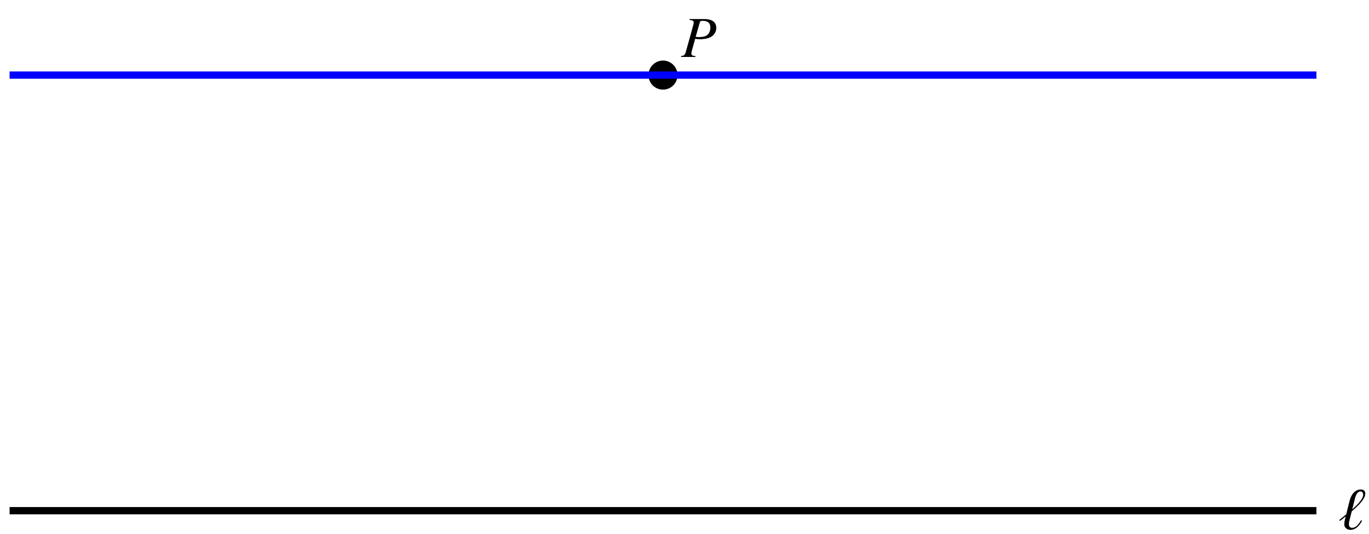
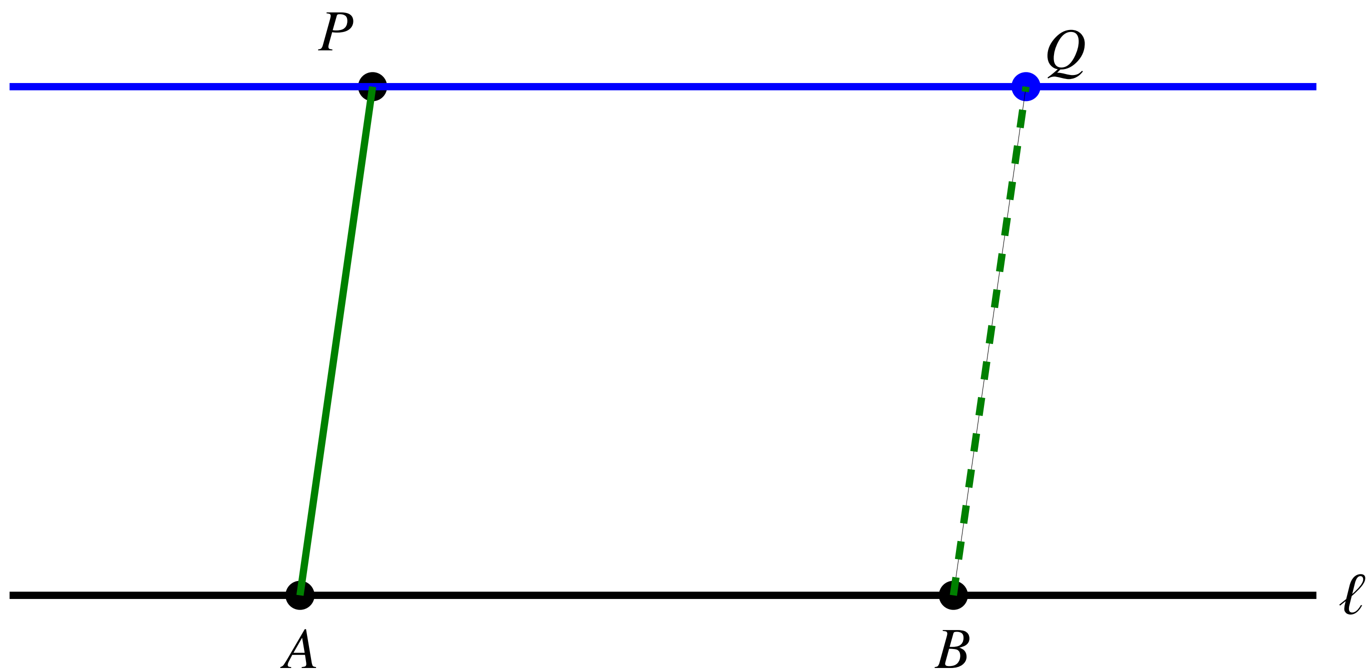
Lecture 5: More on Parallelograms

Recap of previous lecture

New construction: Given a point P and a line ℓ, we can construct the line through P parallel to ℓ.



By marking points on the lines, we can easily construct a parallelogram. For example, if we have points A and B on line ℓ, we can construct a parallelogram by a line parallel to AP through B. So we get a parallelogram $ABQP$.

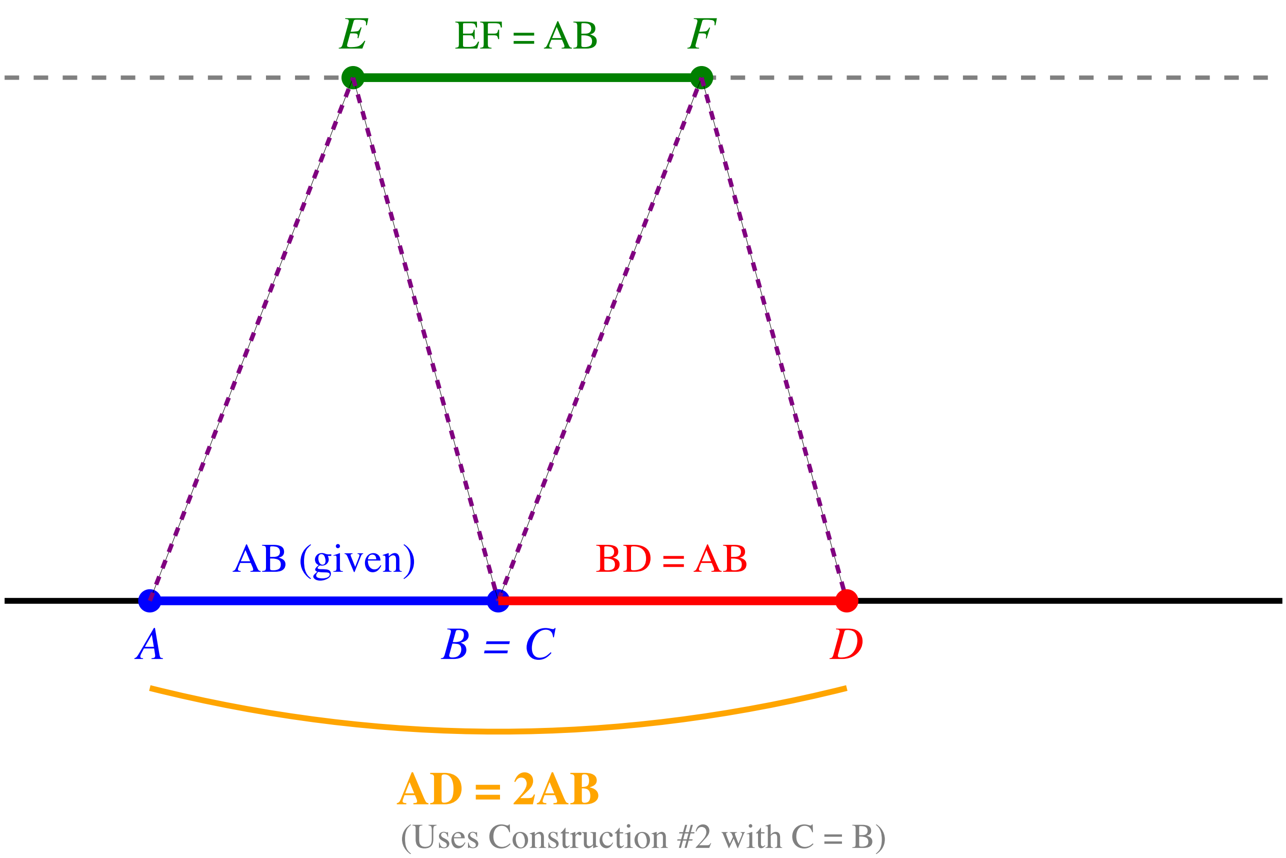


Definition: Opposite edges of a parallelogram are parallel and equal in length.

Double a segment:

Construction #3: Double the Length

Given AB, set C = B, then construct D such that BD = AB



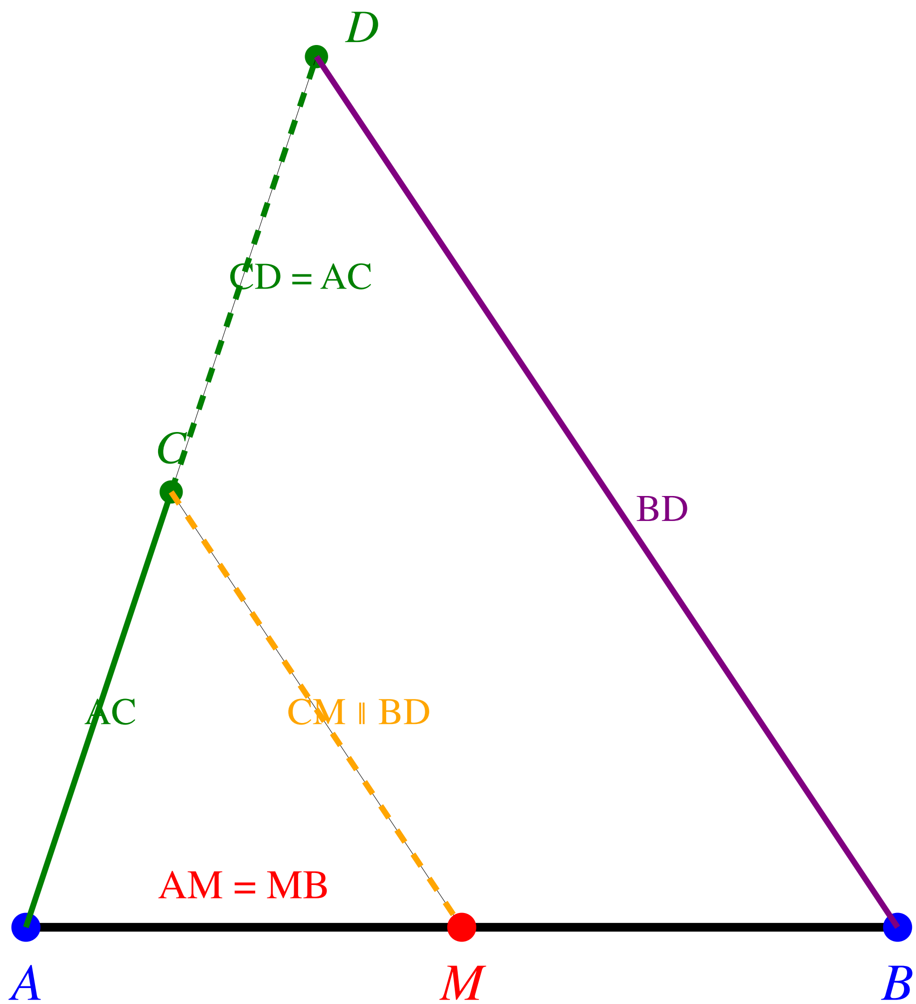
Construction #4: Halving a segment

Given a segment AB, construct its midpoint M such that $AM = MB$.

Steps:

1. Construct another line though A
2. Pick a point C on this line and construct AC.
3. By Construction #3, construct a point D such that $AC = CD$.
4. Construct the line BD.
5. Construct the line through C parallel to BD and let it intersect AB at M.

Construction #4: Halving a Segment



Given AB, construct midpoint M such that AM = MB

Line through C parallel to BD intersects AB at midpoint M

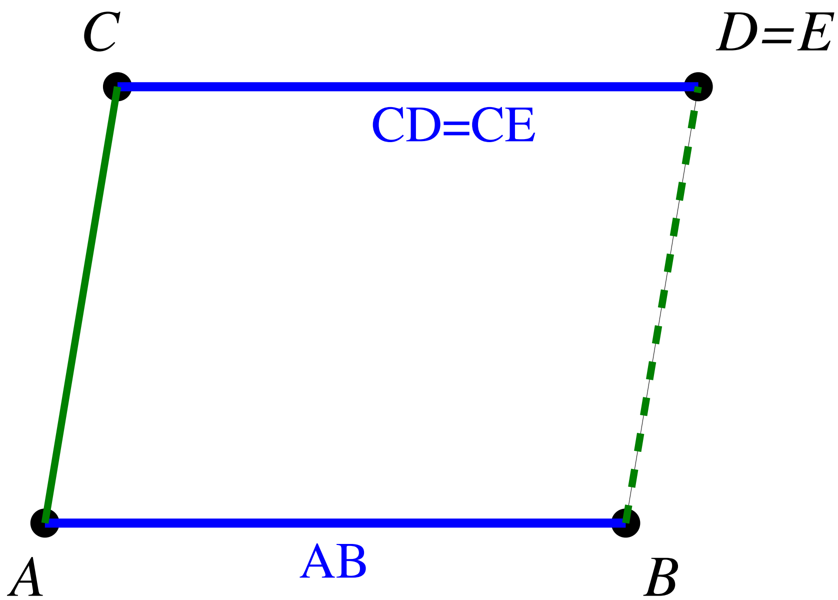
This construcion requires the following key lemma and we will prove it in this lecture.

Key Lemma Equal segments on a line projects to equal segments on another line via parallel lines.

Lemma 1: Given two parallel segments AB and CD such that $AB = CD$, then $ABCD$ is a parallelogram.

Proof:

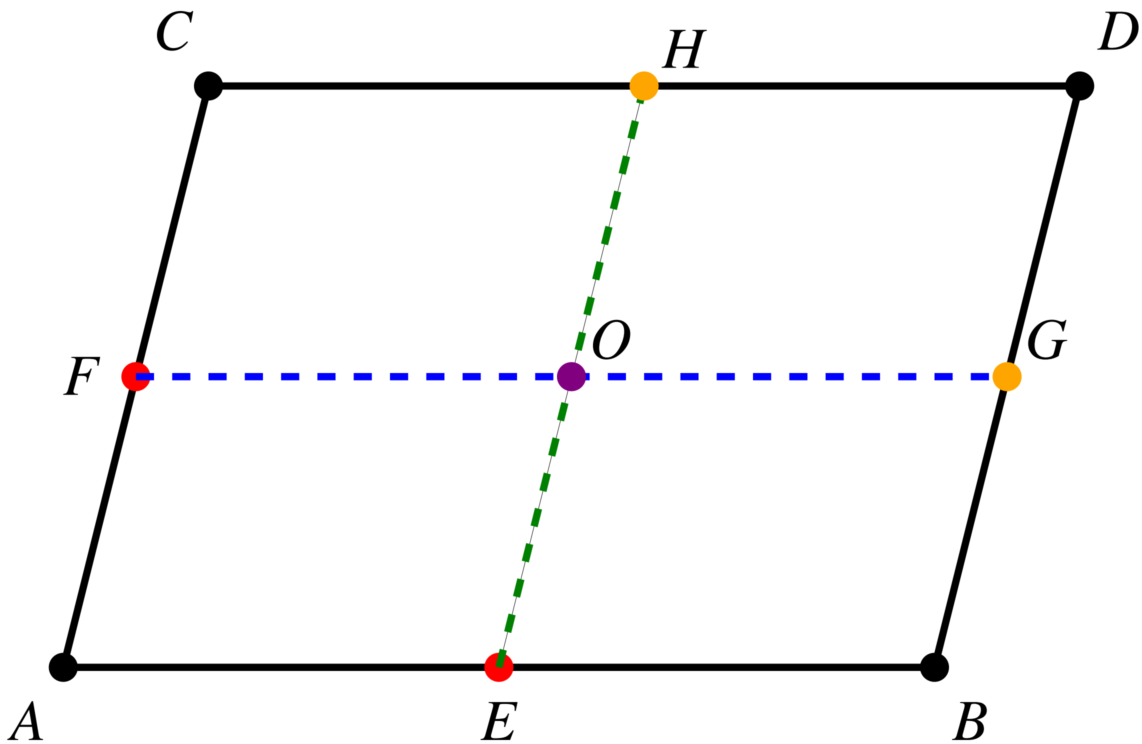
1. Construct the line AC
2. Construct the line through B parallel to AC and let it intersect the line AB at E. So we know $ABCE$ is a parallelogram and $AB = CE$.
3. Now we have $AB = CD = CE$. So $D = E$. Thus, $ABCD$ is a parallelogram.



Thus we have the following **two equivalent definitions of a parallelogram**:

1. two set of parallel lines
2. one pair of parallel lines with equal segments

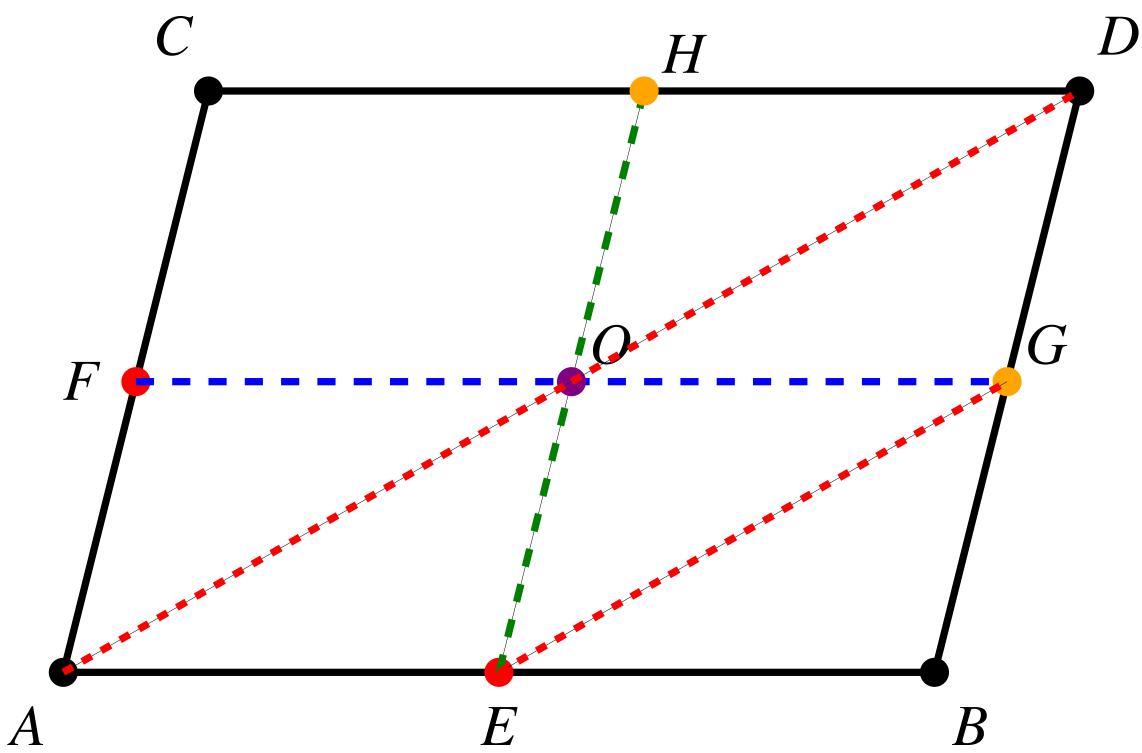
Lemma 2: Given a parallelogram $ABCD$ and mid points E on AB and F on AC, then the line parallel to AB through F and the line parallel to AC through E cut the parallelogram into four equal parallelograms. All horizontal segments are equal and all vertical segments are equal.



Definition: O is called the **center** of the parallelogram $ABCD$ if O is the intersection of FG and EH.

Proof: It is trivial because all small parallelograms are determined by two pairs of parallel lines.

Lemma 3: $AEGO$ and $EGDO$ are parallelograms.



Proof: By Lemma 2, we know that $OG = AE$. Since $OG \parallel AE$, by Lemma 1 we know that $AEGO$ is a parallelogram.

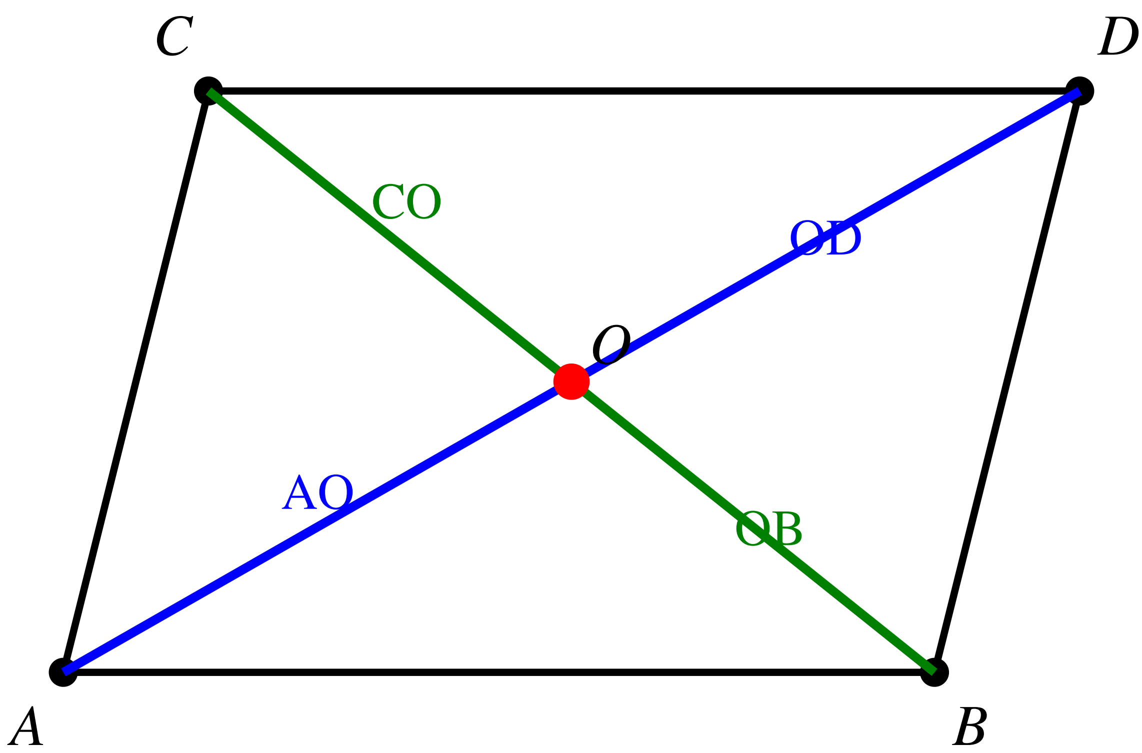
Similarly, by Lemma 2 again, we know that $DG = OE$. Since $DG \parallel OE$, by Lemma 1 we know that $EGDO$ is a parallelogram.

Lemma 4: AO and OD are on the same line as the diagonal AD and $AO = OD$.

Proof: By Lemma 3, we know that $AO \parallel EG$ and $AO = EG$, also $OD \parallel EG$ and $OD = EG$. Thus, $AO = OD$ and they are parallel to EG. By Euclid's fifth postulate, AO and OD are on the same line as AD.

In particular, we prove the following theorem about the center of a parallelogram.

Theorem 1: The center of the parallelogram is on the diagonal and bisects the diagonal, i.e., $AO = OD$ and $CO = OB$.

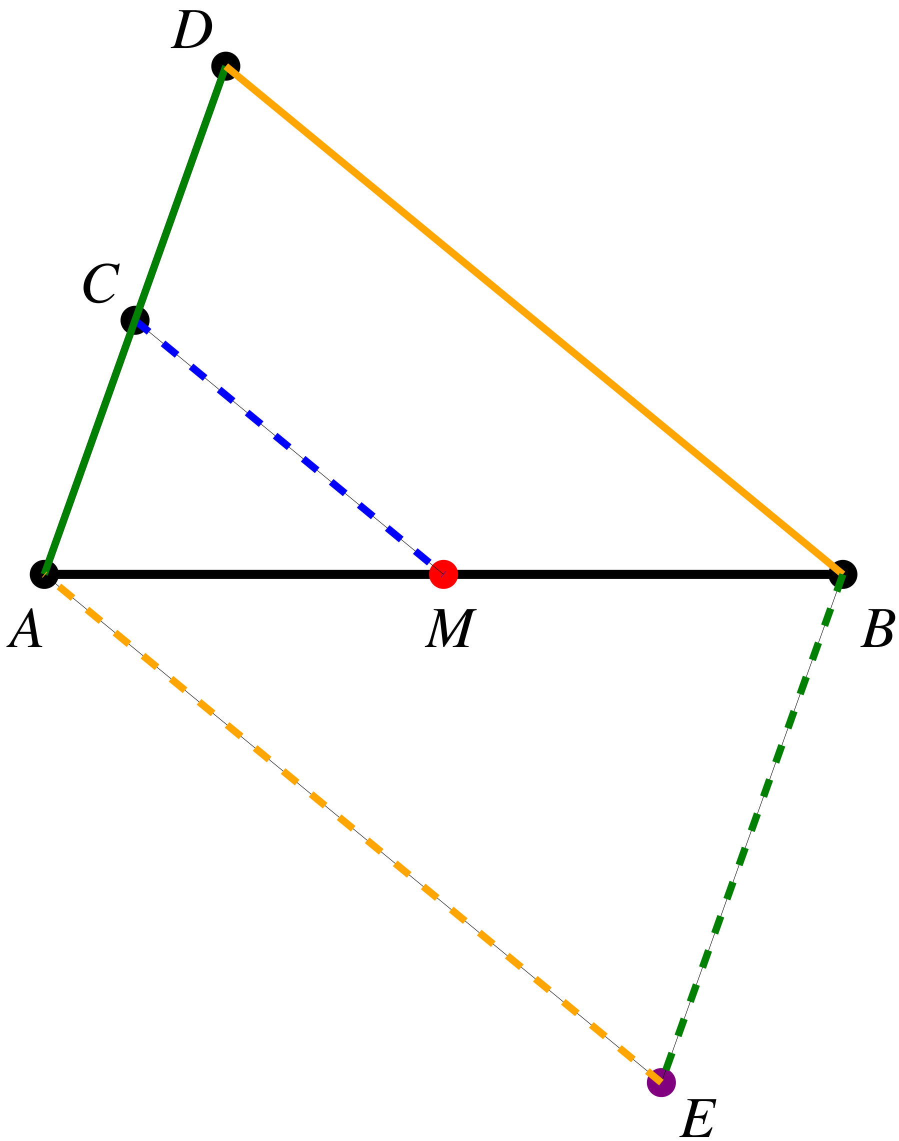


Proof of the key lemma for Construction #4

Now we are ready to prove the key lemma for Construction #4.

Steps:

1. Start with the line segment AB and AC.
2. By Construction #3, construct a point D such that $AC = CD$.
3. Construct the line BD.
4. Construct the line through C parallel to BD and let it intersect AB at M.
5. Construct the parallogram $ADBE$ with $EB \parallel AD$ and $AE \parallel DB$.



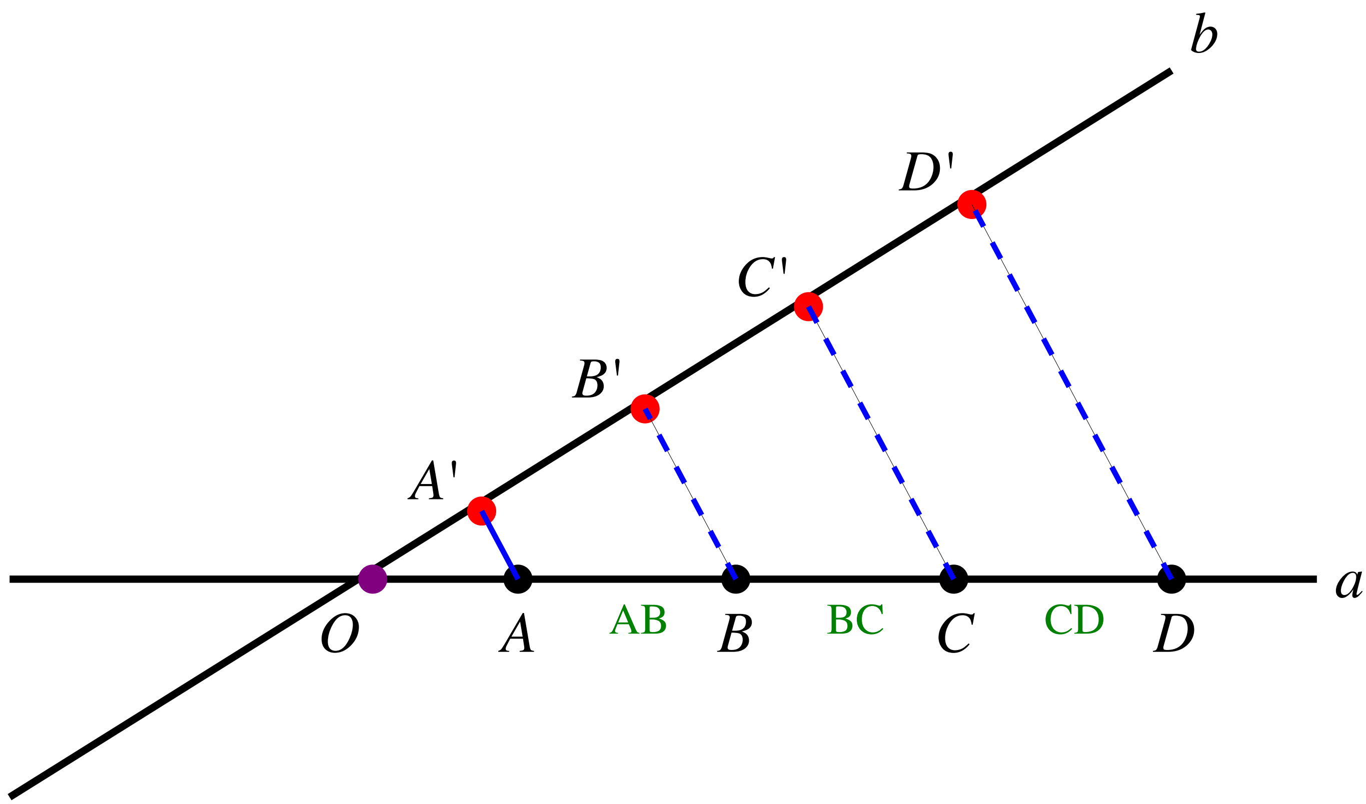
Proof: We know that AB is the diagonal of the parallelogram $ADBE$. We also know that O sits on CM and O sits on AB. So $O = M$. By Theorem 1, the center of the parallelogram is on the diagonal AB and bisects it. Thus, $AM = MB$.

Generalized version of the key lemma

Equal segments project to equal segments between lines

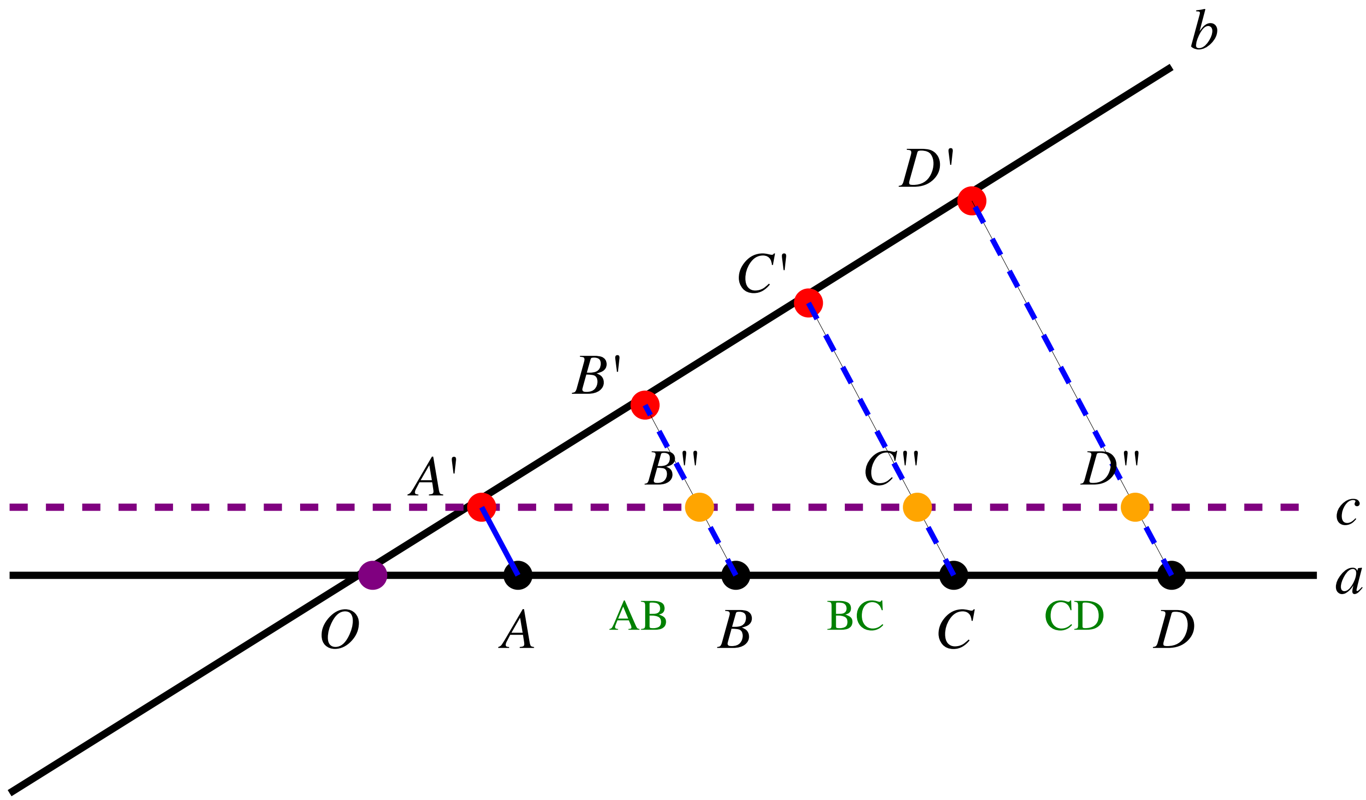
Steps:

1. Construct two lines a, b through O.
2. Construct A, B, C, D on a such that $AB = BC = CD$.
3. Choose any point A' on b.
4. Construct B' such that BB' is parallel to AA'.
5. Construct C' such that CC' is parallel to AA'.
6. Construct D' such that DD' is parallel to AA'.
7. **Then we have $A'B' = B'C' = C'D'$.**



How to prove it?

Construct another line c parallel to a through A'. c intersects BB' at B'', CC' at C'' and DD' at D''.



By parallelogram property, we know that $AB = BC = CD$ implies $A'B'' = B''C'' = C''D''$. Now the key lemma implies that $A'B' = B'C'$ since $A'B'' = B''C''$. Similarly, $B'C' = C'D'$. Thus, $A'B' = B'C' = C'D'$.